

BS-CH101 / BS-CH201 — Chemistry-I

Semester I • 3L:1T:0P • 4 Credits

Module 4 Exam Notes: Free Energy and Chemical Equilibria

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Free Energy and Chemical Equilibria	2
1.1	First and Second Laws of Thermodynamics	2
1.2	Entropy and Free Energy Estimation	2
1.3	Free Energy and Equilibrium Constant	2
1.4	Cell Potential and Nernst Equation	2
1.5	Acid-Base, Redox and Solubility Equilibria	3
1.6	Water Chemistry, Corrosion and Ellingham Diagram	3
2	Derivations, Numericals and Engineering Applications	4
2.1	Deriving the Equilibrium Relation	4
2.2	Entropy Change for Common Processes	4
2.3	Nernst Equation Derivation from Free Energy	4
2.4	Water Hardness and Boiler Problems	5
2.5	Electrochemical Corrosion Mechanism	5
2.6	Ellingham Diagram: Reduction Rule	5
2.7	Temperature Dependence of Spontaneity	6
2.8	pH, Acid Dissociation and Buffer Idea	6
2.9	Solubility Product and Common Ion Effect	7
2.10	Corrosion Prevention Methods	7

1. Free Energy and Chemical Equilibria

Gibbs free energy is the thermodynamic function that predicts spontaneity at constant temperature and pressure. A process is spontaneous when $\Delta G < 0$.

1.1 First and Second Laws of Thermodynamics

The first law states conservation of energy. The second law states that total entropy of the universe increases for a spontaneous process.

Thermodynamic Functions

$$\Delta U = q + w, \quad \Delta H = \Delta U + \Delta(PV)$$

$$\Delta G = \Delta H - T\Delta S$$

At constant temperature and pressure, $\Delta G < 0$ means spontaneous, $\Delta G = 0$ means equilibrium, and $\Delta G > 0$ means non-spontaneous.

1.2 Entropy and Free Energy Estimation

Entropy measures dispersal of energy. For a reversible process,

$$\Delta S = \int \frac{\delta q_{\text{rev}}}{T}$$

Free energy change combines enthalpy and entropy and is useful because it directly predicts useful work and equilibrium direction.

1.3 Free Energy and Equilibrium Constant

For a reaction at non-standard condition,

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium, $\Delta G = 0$ and $Q = K$.

Relation Between Free Energy and Equilibrium Constant

$$\Delta G^\circ = -RT \ln K = -2.303RT \log K$$

1.4 Cell Potential and Nernst Equation

Electrochemical cell work is linked with free energy.

Electrochemical Free Energy

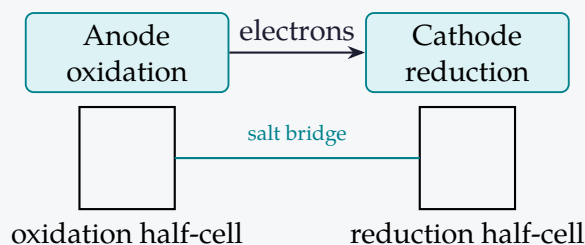
$$\Delta G = -nFE, \quad \Delta G^\circ = -nFE^\circ$$

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298 K:

$$E = E^\circ - \frac{0.0591}{n} \log Q$$

Galvanic Cell and Electron Flow



1.5 Acid-Base, Redox and Solubility Equilibria

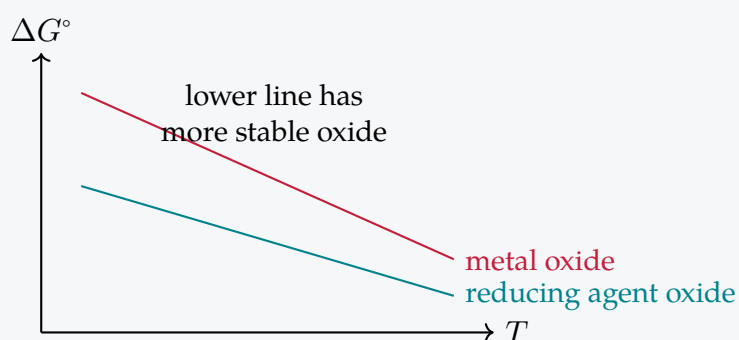
The same free-energy idea applies to acid-base ionization, redox reactions, and precipitation.

Equilibrium	Constant	Use
Acid-base	K_a, K_b	pH and strength of acid/base
Redox	E°, K	Feasibility of electron transfer
Solubility	K_{sp}	Precipitation and ionic product

1.6 Water Chemistry, Corrosion and Ellingham Diagram

Water chemistry includes hardness, alkalinity, pH, and dissolved oxygen. Corrosion is an electrochemical oxidation of metal. Ellingham diagrams plot ΔG° versus temperature and help choose reducing agents in metallurgy.

Ellingham Diagram Concept



Exam Points

- Always state sign of ΔG for spontaneity.
- For equilibrium questions, start from $\Delta G = \Delta G^\circ + RT \ln Q$.
- For electrochemistry, connect ΔG , E , and Nernst equation.
- For Ellingham diagrams, lower line means more stable oxide.

2. Derivations, Numericals and Engineering Applications

2.1 Deriving the Equilibrium Relation

For any reaction:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium, the reaction has no net driving force, so:

$$\Delta G = 0, \quad Q = K$$

Therefore:

$$0 = \Delta G^\circ + RT \ln K$$

$$\Delta G^\circ = -RT \ln K$$

Using $\ln K = 2.303 \log K$:

$$\Delta G^\circ = -2.303RT \log K$$

Solved Example: Equilibrium Constant from Free Energy

At 298 K, suppose $\Delta G^\circ = -11.4 \text{ kJ mol}^{-1}$.

$$\log K = \frac{-\Delta G^\circ}{2.303RT}$$

$$\log K = \frac{11400}{2.303(8.314)(298)} \approx 2.00$$

$$K \approx 10^2 = 100$$

Since $K > 1$, products are favoured.

2.2 Entropy Change for Common Processes

Entropy increases when randomness, volume, or number of gas molecules increases.

Process	Entropy change	Reason
Melting of solid	Positive	Liquid has more molecular freedom
Vaporization	Strongly positive	Gas has much greater randomness
Compression of gas	Negative	Accessible volume decreases
Dissolution of salt	Usually positive	Ions disperse in solution
Crystallization	Negative	Ordered solid forms

2.3 Nernst Equation Derivation from Free Energy

Electrical work from a cell is:

$$\Delta G = -nFE$$

At standard condition:

$$\Delta G^\circ = -nFE^\circ$$

Substitute both in:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$-nFE = -nFE^\circ + RT \ln Q$$

Dividing by $-nF$:

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298 K:

$$E = E^\circ - \frac{0.0591}{n} \log Q$$

Solved Example: Cell Potential

For a cell with $E^\circ = 1.10$ V, $n = 2$, and $Q = 10$:

$$E = 1.10 - \frac{0.0591}{2} \log 10$$

$$E = 1.10 - 0.02955 = 1.07045 \text{ V}$$

2.4 Water Hardness and Boiler Problems

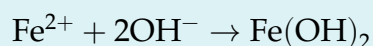
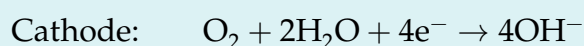
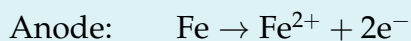
Hard water contains dissolved calcium and magnesium salts. Temporary hardness is due to bicarbonates; permanent hardness is due to chlorides and sulphates.

Problem	Cause	Effect in engineering systems
Scale formation	Deposition of salts like CaCO_3	Poor heat transfer and tube overheating
Sludge formation	Loose precipitates in boiler water	Choking and reduced efficiency
Priming	Carryover of water droplets with steam	Wet steam and turbine damage
Foaming	Stable bubbles due to impurities	Irregular boiler operation
Caustic embrittlement	Concentrated alkali in cracks	Boiler metal failure

2.5 Electrochemical Corrosion Mechanism

In iron corrosion, anodic and cathodic regions form on the metal surface.

Rusting Half-Reactions



Further oxidation gives hydrated ferric oxide, commonly called rust.

2.6 Ellingham Diagram: Reduction Rule

In metallurgy, a metal oxide can be reduced by an element whose oxide line lies below it in the Ellingham diagram. This means the reducing agent forms a more stable oxide and

makes the overall ΔG° negative.

Answer-Writing Pattern for Ellingham Problems

1. Identify the oxide to be reduced.
2. Locate its ΔG° line at the given temperature.
3. Find a reducing agent whose oxide line is lower.
4. State that the net free energy change becomes negative.

2.7 Temperature Dependence of Spontaneity

The sign of $\Delta G = \Delta H - T\Delta S$ depends on both enthalpy and entropy. This table is a common 5-mark answer.

Case	Sign condition	Spontaneity
Exothermic, entropy increases	$\Delta H < 0, \Delta S > 0$	Spontaneous at all temperatures
Endothermic, entropy decreases	$\Delta H > 0, \Delta S < 0$	Non-spontaneous at all temperatures
Exothermic, entropy decreases	$\Delta H < 0, \Delta S < 0$	Spontaneous at low temperature
Endothermic, entropy increases	$\Delta H > 0, \Delta S > 0$	Spontaneous at high temperature

Solved Example: Temperature of Spontaneity

For a reaction, $\Delta H = 40 \text{ kJ mol}^{-1}$ and $\Delta S = 100 \text{ J mol}^{-1} \text{ K}^{-1}$. Convert entropy:

$$\Delta S = 0.100 \text{ kJ mol}^{-1} \text{ K}^{-1}$$

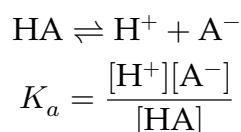
At equilibrium, $\Delta G = 0$:

$$T = \frac{\Delta H}{\Delta S} = \frac{40}{0.100} = 400 \text{ K}$$

Since both ΔH and ΔS are positive, the reaction is spontaneous above 400 K.

2.8 pH, Acid Dissociation and Buffer Idea

For a weak acid:



For a buffer made of weak acid and its salt:

$$\text{pH} = \text{p}K_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Solved Example: Buffer pH

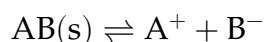
If $pK_a = 4.75$, $[\text{salt}] = 0.20 \text{ M}$ and $[\text{acid}] = 0.10 \text{ M}$:

$$\text{pH} = 4.75 + \log \frac{0.20}{0.10}$$

$$\text{pH} = 4.75 + \log 2 = 4.75 + 0.301 = 5.051$$

2.9 Solubility Product and Common Ion Effect

For a sparingly soluble salt AB:



$$K_{sp} = [\text{A}^+][\text{B}^-]$$

Precipitation occurs when ionic product exceeds K_{sp} .

Condition	Meaning	Observation
Ionic product $< K_{sp}$	Unsaturated solution	No precipitation
Ionic product $= K_{sp}$	Saturated solution	Equilibrium
Ionic product $> K_{sp}$	Supersaturated with respect to salt	Precipitation occurs

2.10 Corrosion Prevention Methods

Corrosion can be reduced by blocking contact with environment or by making the metal cathodic.

Method	Principle	Example
Painting / coating	Barrier between metal and environment	Paint, oil, polymer coating
Galvanization	Zinc corrodes sacrificially before iron	Zn-coated iron sheets
Cathodic protection	Metal to be protected is made cathode	Pipelines, ship hulls
Alloying	Forms corrosion-resistant surface film	Stainless steel
Inhibitors	Reduce anodic or cathodic reaction rate	Boiler-water additives

Previous-Year Style Questions

1. Derive $\Delta G^\circ = -RT \ln K$ and explain its significance.
2. Derive Nernst equation and solve a cell-potential numerical.
3. Explain water hardness, scale formation and boiler troubles.

4. Describe electrochemical corrosion of iron and prevention methods.
5. Explain Ellingham diagram and its use in metallurgy.

Quick Revision — Module 4

Free Energy and Chemical Equilibria | BS-CH101

Key Formulas at a Glance

Formula / Rule	Meaning
$\Delta U = q + w$	First law of thermodynamics.
$\Delta G = \Delta H - T\Delta S$	Gibbs free energy relation.
$\Delta G = \Delta G^\circ + RT \ln Q$	Free energy at non-standard condition.
$\Delta G^\circ = -RT \ln K$	Relation between free energy and equilibrium constant.
$\Delta G = -nFE$	Cell free energy and cell potential.
$E = E^\circ - \frac{RT}{nF} \ln Q$	Nernst equation.
$K_{sp} = [M^{m+}]^n [X^{n-}]^m$	Solubility product expression.

One-Line Definitions

- **Entropy:** Measure of energy dispersal or randomness.
- **Gibbs free energy:** Energy available for useful work at constant temperature and pressure.
- **Equilibrium constant:** Ratio of product and reactant activities at equilibrium.
- **Nernst equation:** Relation between electrode potential and concentration.
- **Corrosion:** Electrochemical deterioration of a metal.
- **Ellingham diagram:** Plot of ΔG° for oxide formation versus temperature.

Frequently Asked / Viva Questions

Q1. State the first law of thermodynamics.

Energy cannot be created or destroyed; $\Delta U = q + w$.

Q2. What is the criterion of spontaneity using Gibbs free energy?

$\Delta G < 0$ spontaneous, $\Delta G = 0$ equilibrium, $\Delta G > 0$ non-spontaneous.

Q3. Write Gibbs free energy equation.

$\Delta G = \Delta H - T\Delta S$.

Q4. Relate ΔG° and equilibrium constant.

$\Delta G^\circ = -RT \ln K$.

Q5. How is cell emf related to free energy?

$\Delta G = -nFE$.

Frequently Asked / Viva Questions (contd.)

Q6. State the Nernst equation at 298 K.

$$E = E^{\circ} - (0.0591/n) \log Q.$$

Q7. What is solubility product?

It is the product of ion concentrations in a saturated solution, each raised to stoichiometric power.

Q8. Why does corrosion occur electrochemically?

Metal oxidation and reduction of oxygen or hydrogen occur at different surface sites.

Q9. What does an Ellingham diagram show?

It shows temperature dependence of standard free energy of oxide formation.

Q10. How is a reducing agent selected from Ellingham diagram?

A reducing agent whose oxide line lies below the metal oxide line can reduce that oxide.

Summary Reference Table

Area	Key relation	Exam focus
Thermodynamics	$\Delta G = \Delta H - T\Delta S$	Spontaneity and equilibrium
Chemical equilibrium	$\Delta G^{\circ} = -RT \ln K$	Direction and extent of reaction
Electrochemistry	$\Delta G = -nFE$	Cell potential and Nernst equation
Metallurgy	Ellingham diagram	Reduction feasibility